



Document Control

Document Title: Job Description: Integrations Engineer

Document Owner: Information Security / Management

Classification: Internal

Date: 2026-04-30

Job Description: Systems Integrations Engineer (Real-time system)

Who are we?

At 3Dtracking, we build technology that makes complex fleet operations easier—easier to manage, easier to scale, and easier to trust.

With over 20 years in the industry and a platform used in 150+ countries, we are a globally recognized leader in white-labeled telematics. Our software powers mission-critical fleet operations for thousands of service providers across six continents. But what sets us apart isn't just the tech, it's how we support our customers, evolve with them and help them succeed in a complex industry.

Role Overview

We are looking for a hands-on Systems Integrations Engineer to design, build and support high-performance data integrations between telematics devices, the 3Dtracking platform and third-party systems.

This role is heavily focused on:

- TCP device ingestion (high-volume, real-time data)
- Backend processing pipelines
- API and system integrations
- Data reliability and performance at scale.

You'll work closely with internal engineering teams and external partners to deliver robust, production-grade integrations.

Key Responsibilities

1. Device & Data Ingestion (Core Focus)

- Build and maintain TCP servers/services for ingesting telemetry data from tracking devices
- Parse, validate and normalise high-volume incoming data streams
- Handle unreliable networks, retries and malformed device payloads
- Optimise ingestion pipelines for throughput, latency and fault tolerance.

2. Integration Development

- Design and build integrations with:
 - Fleet platforms
 - ERPs / CRMs
 - Partner systems
- Develop REST APIs, webhooks, and event-driven integrations
- Implement secure authentication (OAuth2, API keys, certificates)
- Build reusable integration components and frameworks.

3. Back-end Engineering

- Develop services primarily in C# (.NET)
- Work with MS SQL Server for high-performance data storage and querying
- Design and optimise schemas for time-series / telemetry data
- Write efficient queries and improve database performance.

4. Eventing & Messaging

- Work with message queues / brokers (e.g. RabbitMQ, Kafka)
- Implement asynchronous processing pipelines
- Ensure reliable delivery and processing of integration events.

5. Reliability & Monitoring

- Implement logging, monitoring, and alerting for integration services
- Troubleshoot production issues across device, network and system layers
- Build retry, queuing and fallback mechanisms
- Ensure high availability of ingestion and integration services.

6. Partner & Customer Integration Support

- Assist partners during technical onboarding
- Participate in integration design sessions
- Troubleshoot real-world integration issues (devices, APIs, data mismatches)
- Provide clear technical guidance to internal and external teams.

7. Documentation

- Document:
 - Device protocols
 - Integration flows
 - APIs and data mappings
- Maintain confluence and troubleshooting guides.

Required Skills & Experience

- 4+ years' experience in backend/infrastructure development or systems integration
- Strong experience with **C# / .NET**
- Solid experience with **MS SQL Server** (query optimisation, schema design)
- Experience building and maintaining **TCP/IP-based services**
- Strong understanding of **real-time data processing systems**
- Experience with **REST APIs, webhooks, and distributed systems**
- Experience with **message queues** (RabbitMQ, Kafka, etc.)
- Understanding of authentication and security (OAuth2, JWT, API keys).

Advantageous

- Experience in telematics, IoT or GPS tracking systems
- Experience handling high-volume event/data ingestion systems
- Exposure to Azure or cloud-hosted environments
- Experience working with binary protocols or custom device formats.

What we're Looking For

- Strong systems thinker (not just feature developer)
- Comfortable debugging across network, device and back-end layers
- Able to handle messy real-world data and edge cases
- Practical, delivery-focused mindset
- Comfortable working in a fast-moving, scaling environment.

What Success Looks Like

- Stable, reliable device ingestion pipelines
- Scalable integrations that handle increasing device volumes
- Reduced data loss and faster issue resolution
- Faster onboarding of new devices and partners
- Clear, maintainable integration architecture

Why Join Us

- Work on real-world, large-scale telemetry systems
- Solve complex data and integration challenges
- Play a key role in scaling a global telematics platform
- Collaborative and technically strong team

Details

- Location: Cape Town (Hybrid)
- Type: Permanent, Full-Time
- Department: Engineering
- Reports to: Infrastructure Lead